Selection of building site










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Site selection is the first and most important step



For every site there is an ideal use,

For every use there is an ideal site.

Building classified based upon occupancy

- Residential building
  - ✓ Single family houses
  - ✓ Multi family dwellings
  - ✓ High-rise apartments
- Institutional Construction
  - ✓ Schools and universities
  - ✓ Medical clinics and hospitals
  - ✓ Recreational facilities and sports stadiums
- Assembly building
  - Hotels, convention centers, and theaters
- Storage building
  - Warehouses and light manufacturing
- Commercial buildings
  - ✓ Retail stores and shopping centers
  - ✓ Office buildings (single story to sky scrappers)
- Industrial buildings
  - ✓ Petroleum refineries
  - ✓ Steel mills & aluminum plants
  - ✓ Chemical processing plants
  - ✓ Fossil fuel & nuclear power plants
  - ✓ Other heavy manufacturing facilities
- Heavy Construction

Heavy Construction

- Railroads / Urban Transit Systems
- Highway & Bridges
- Port & harbor structures
- Tunnels and Dams
- Canals
- Pipelines
- Airports
- systems
- Landfills
- Power & communication networks
- Sewer / Water treatment & distribution





Types based on materials / execution

- Solid or ‘traditional’ construction
- Structural steel frame construction
- Reinforced concrete construction
- Modular system construction
- Portal frame construction
- Composite construction.

One & Two Family



Residential Type “3” building



Multiple Residence



Commercial and Residential

Residential Styles


ART DECO


BUNGALOW


CAPE COD


COLONIAL


CONTEMPORARY


CRAFTSMAN


CREOLE


DUTCH COLONIAL


FEDERAL


FRENCH PROVINCIAL


GEORGIAN


GOTHIC REVIVAL


GREEK REVIVAL


INTERNATIONAL


ITALIANATE


MONTERREY


NATIONAL


NEOCLASSICAL


PRAIRIE


PUEBLO


QUEEN ANNE


RANCH


REGENCY


SALTBOX


SECOND EMPIRE


SHED


SHINGLE


SHOTGUN


SPANISH ECLECTIC

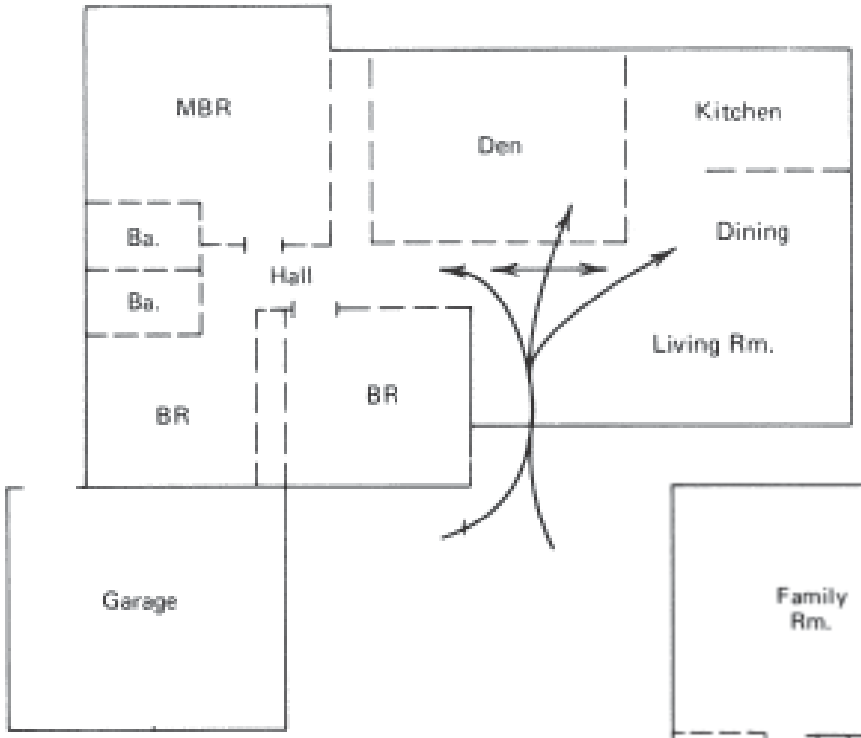

SPLIT LEVEL


STICK


TUDOR

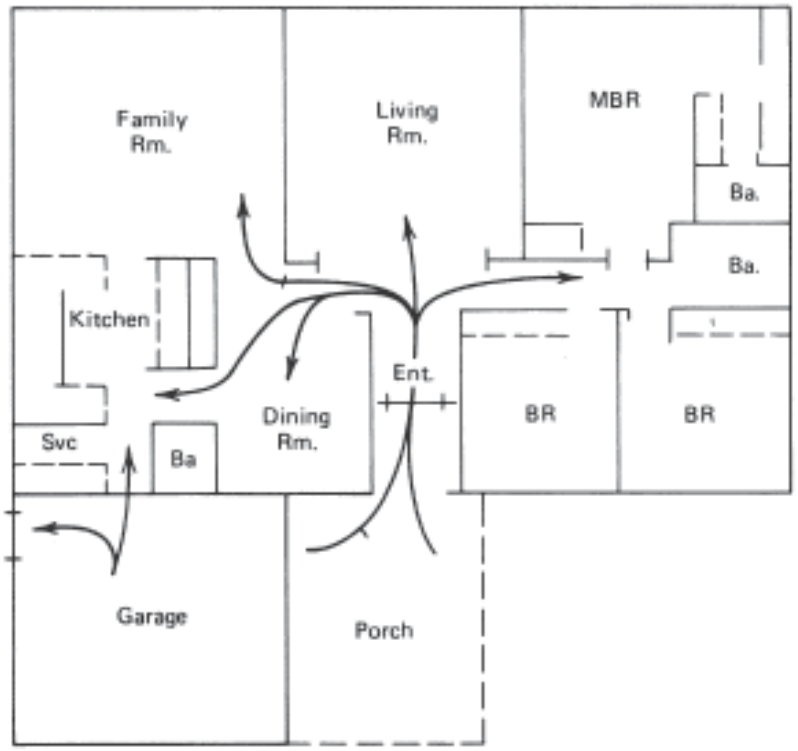

VICTORIAN

Different floor plans to choose



House A has:

- Some lack of visual and auditory privacy
- Notable traffic congestion
- Inconvenient access to garage
- Lack of cross-ventilation in bedroom adjacent to garage



House B has:

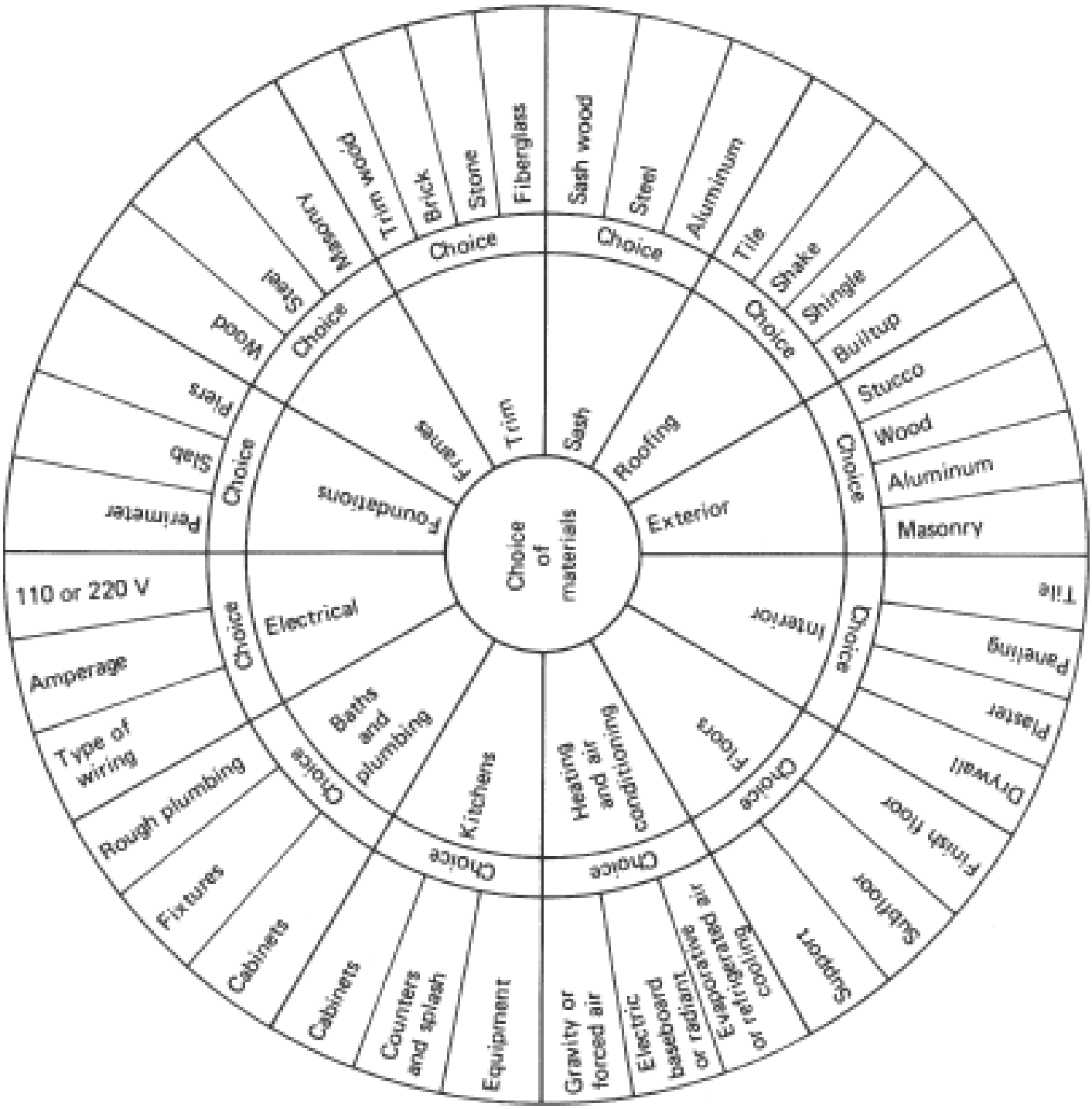
- Good to excellent visual and auditory privacy
- Very good traffic flow
- Convenient service area and garage location
- Evidence of good cross-ventilation

Residential buildings – Inside access

- Orientation
- Aspect
- Privacy
- Grouping
- Circulation
- Furnishing
- Sanitation
- Ventilation
- Flexibility
- Landscaping
- Economy



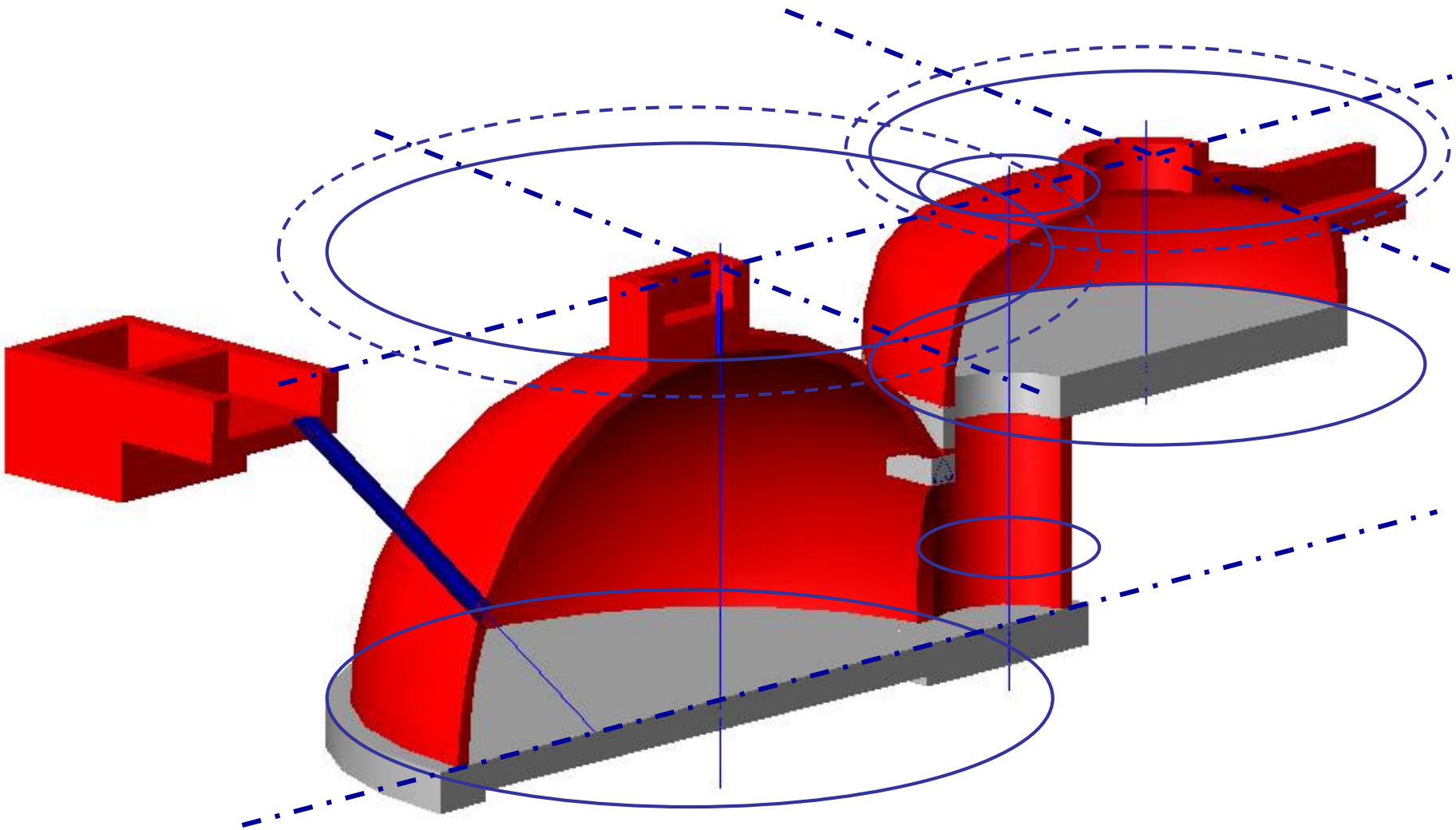
CHOICE OF MATERIALS



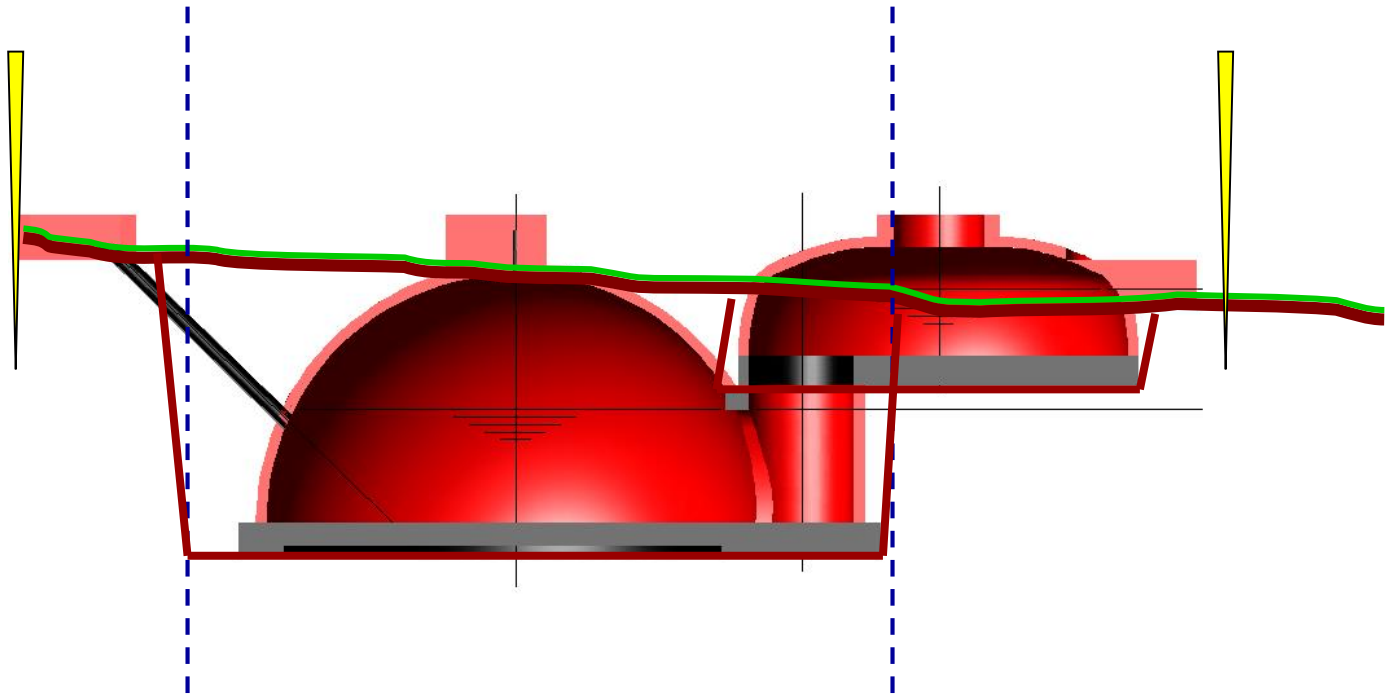
Setting out



Setting out for lateral construction



Space for external projections



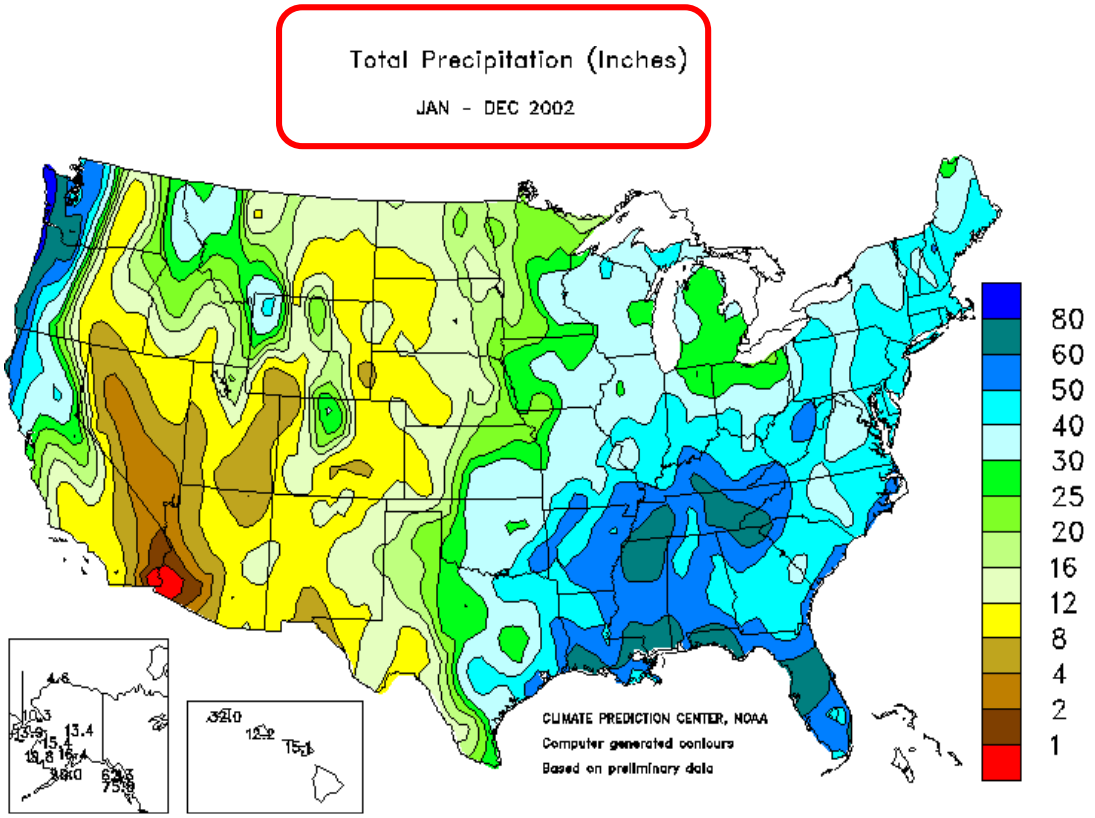
Site selection is the first and most important step in designing a house

Climate	Soil type
Required area	Water
Orientation	Utilities
Prevailing wind direction	Access
Elevation	Future expansion
Slope	Off-farm factors



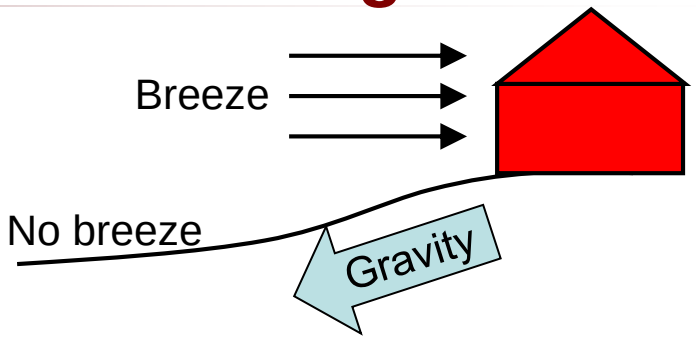
Climate refers to the average weather conditions of a given location.

- Temperature
  - Extremes – Hot / Cold
  - Average
- Humidity – High / low
- **Precipitation**
  - Rainfall
  - Snowfall



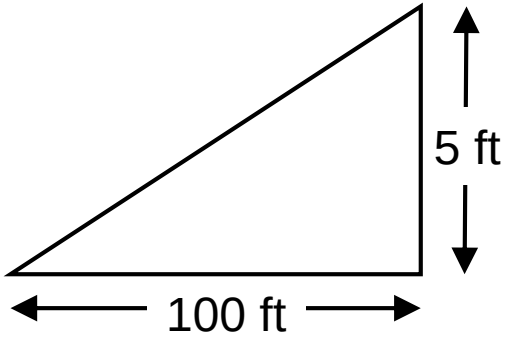
Elevation – take advantage of natural ridges when considering site selection.

- Maximize cooling breezes
- Maximize sunshine
- Provide gravity flow
  - ✓ Flushing systems



Slope can have a great impact on the cost of any building project.

- Too steep
  - Requires extensive grading
  - Expensive
- Too flat
  - Cannot use gravity
  - Expensive
- Provide:
  - 2 to 6% slope for drainage



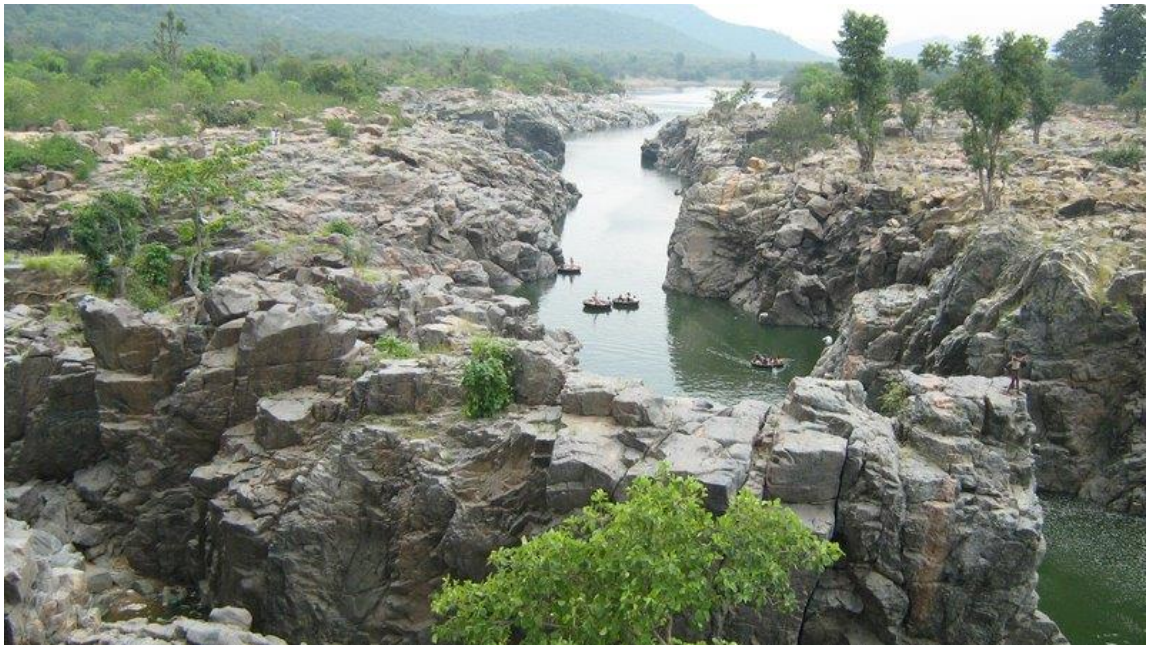
Slope of 5% (0.05).

Soil type should be determined

- Electric
- Fuel / Gas
- Telephone



- Structural support
- Surface & sub-surface drainage
- Depth to bedrock
- Depth to water table
- Erodibility



Site selection

- Location- Is it in an area where there are a lot of potential user/ customers.
- The site should be located in a fully developed or a fast developing area with transporting facilities.
- Consider when selecting a site for future construction.
- Size and shape- Is the site large enough for the planned construction. If not, can the adjoining side be acquired.
- The site should not be irregular in shape or having any sharp corners.
- Maximum benefit is derived from nature, if the buildings are properly oriented.
- The orientation is also dependent on the layout of existing roads and buildings in the locality.
- Cost – Is the price of the site reasonable and affordable?

Site selection for buildings (contd…)

- The site should be abandoned under adverse circumstances such as (a) unhealthy, noisy or crowded localities; (c) Reclaimed soil or water lagged areas, subject to submergence or settlement; and (d) Industrial vicinities having smoke and obnoxious odour.
- Doors and windows should be positioned so as to get maximum lighting, breeze and view of the landscape.
- Privacy inside the house is achieved by proper placing of windows.
- Privacy outside the house is achieved by growing trees and creepers.
- The doors and windows should be placed opposite to each other to ensure maximum ventilation.

Site selection for buildings

- The plot should be in a locality where the various facilities like (a) Community services such as police and fire protection, clearing of waste and street cleaning; (b) Utility services such as water supply, gas, electricity and drainage; (c) Amenities such as schools, hospitals, libraries, recreation, telephone etc.; (d) shopping facilities and (e) means of transport; are available.
- The legal and financial aspects should be given due consideration before the purchase of the plot.
- The proportions of plot to be built-up, vacant spaces to be left in front and sides, heights of buildings, etc., should be in accordance with local byelaws.
- Zoning – Will zoning laws permit the type of planned structure to be built there.

Zoning Laws

- No matter what type of structure is desired or who desires it, it must meet the requirements and standards set up by the community in which it is to be built.
- Zoning laws tell what kind of structures can be built in certain areas.
- May specify such things as:
  - ✓ Maximum property size
  - ✓ Maximum height of buildings
  - ✓ Number of families that can occupy a home.
  - ✓ Number of parking spaces
  - ✓ Distance structures must be from the properties boundary lines.

IMPROVEMENT CHARACTERISTICS

- Exterior and Interior Features



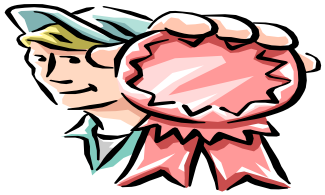
- Appeal and Marketability



- Construction Quality



- Physical Condition



- Effective Age
  - ✓ Relative age of a structure considering its physical condition and marketability.



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